

Clinical assessment: moving from rote to rigorous

by CHAD COOK

Listen: the patient is telling you the diagnosis.

Sir William Osler, Canadian physician (1849–1919)

Clinical assessment is every top clinician's foundation stone and it is a complex skill. Its three goals are to:

1. determine the diagnosis of the patient
2. identify appropriate treatment mechanisms specific to that condition, and
3. build a professional, interpersonal relationship with the patient to guide efficient management.

Differential diagnosis is used to identify the pertinent diagnosis (condition) of the patient and is the focus of this chapter. This chapter should be read together with Chapter 15.

Differential diagnosis is a systematic process used to identify the proper diagnosis from a competing set of possible diagnoses. The diagnostic process involves identifying or determining the aetiology of a disease or condition through evaluation of patient history, physical examination, and review of laboratory data or diagnostic imaging; and the subsequent descriptive title of that finding.¹

WHY IS DIFFERENTIAL DIAGNOSIS IMPORTANT?

Failure to correctly identify an appropriate diagnosis can lead to negative patient-reported outcomes,² delays in appropriate treatment,¹ and unnecessary healthcare costs.³

What makes diagnostic tests inaccurate?

- failure to order appropriate tests (58% of instances)
- inadequate history/physical examination (42% of instances)
- incorrect test interpretations (37% of instances).⁴

With respect to medical errors, diagnostic errors are the most commonly recorded type (29% of all errors) and account for the highest proportion of total payments (35%).⁵

PRACTICE PEARL

Differential diagnostic errors result in death or disability almost twice as often as other error categories.

Although the exact prevalence of diagnostic error remains unknown, data from autopsy series spanning several decades conservatively and consistently reveal error rates of 10–15%.⁶

DIFFERENTIAL DIAGNOSIS: A THREE-STEP PROCESS

Regardless of treatment environment, most clinicians follow a three-step questioning process during diagnostic assessment.⁷

1. The first question of diagnosis involves the query of whether the patient's symptoms or emergent injury are reflective of a visceral disorder or a serious or potentially life-threatening illness. It is critical to be able to differentiate patients with symptoms that arise from a potentially life-threatening pathology or a non-mechanical disorder (i.e. referred pain).
2. The second question of diagnosis involves determining from where is the patient's pain arising? This step has three substeps that involve: a. ruling out a location; b. ruling in a location but not knowing the tissue-related structure; and c. confirming the tissue-related structure that is causal. Although it is assumed that one can make an accurate tissue-related diagnosis, it is well known that differentiating tissue in the low back, shoulder, abdomen and hip is very challenging and it is not uncommon to see clinicians treat these areas without full knowledge of the tissue of origin. We will discuss this further at the end of this chapter.

3. The third question of diagnosis involves determining what has gone wrong with this person as a whole that would cause the pain experience to develop and persist.⁷ This stage involves careful exploration of the social, psychosocial and socioeconomic contextual elements. This element is outlined in Chapter 17 (treatment) and given more detailed attention in Chapters 23 (neck pain) and 29 (low back pain).

HOW TO CALCULATE AN ACCURATE DIAGNOSIS

Diagnostic accuracy can be evaluated through a common set of metrics, metrics that are universally used regardless of testing type (e.g. imaging [Chapter 15], clinical testing, laboratory testing).

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Absolutely essential metrics that every clinician MUST understand include reliability, sensitivity and specificity, positive and negative predictive values, and positive and negative likelihood ratios (LR+ and LR–, respectively).

Reliability

In diagnostic terms, reliability involves the ability of a test to consistently identify a similar finding when retested (e.g. dynamometer testing for strength), or when used by a different clinician, or the level of consistent agreement among two or more clinicians when a particular test which requires interpretation is used (e.g. Lachman's test). Reliability is a required characteristic of diagnostic testing and is generally scored by 0–1 if one evaluates a numerically scored test such as a goniometer, or 0–1 if one assesses agreement among clinicians (higher scores reflecting greater agreement).

Sensitivity and specificity

Sensitivity and specificity are internal measures (meaning they are relatively independent of the sample) that are used in two distinct populations: 1. the disease of interest or injured population (sensitivity); and 2. a competing disease or non-injured population (specificity). Sensitivity is the percentage of people who test positive for a specific disease among a group of people who have the disease and this value is generally scored from 0%–100%. Higher values have a greater ability to accurately identify those *who have the disease of interest*. Specificity is the percentage of people who test negative for a specific disease among a group of people who do not have the disease and this value is also scored from 0%–100%. Higher values are more accurate at identifying those *who do not have the disease of interest*.

To recap, sensitivity and specificity capture values from two distinct populations: 1. the disease of interest or injured population (sensitivity); and 2. a competing disease or non-injured population (specificity). Figure 14.1 represents the conditions of interest for calculating sensitivity and specificity. Sensitivity only reflects those who have a 'true positive' and a 'false negative' test finding. Specificity only represents those who have a 'true negative' and a 'false positive' test finding. As one can see, each value only represents (potentially) one-half of the diagnostic story, and using these values independently is a mistake commonly made among clinicians.

PRACTICE PEARL

The assumptions of SP in ('high specificity rules in') and SN out ('high sensitivity rules out') are erroneously based on these assumptions.

Unfortunately, these outdated assumptions are misguided and cannot be advocated for clinical practice, primarily because routinely used tests have very high sensitivity and very low specificity (and vice versa), eliminating their capacity to discriminate conditions.

Positive and negative predictive values

Predictive values reflect the clinical setting. They are relevant when one asks the clinical question, 'The clinical screening test is positive. What is the likelihood that this patient truly has the condition?' Positive predictive value

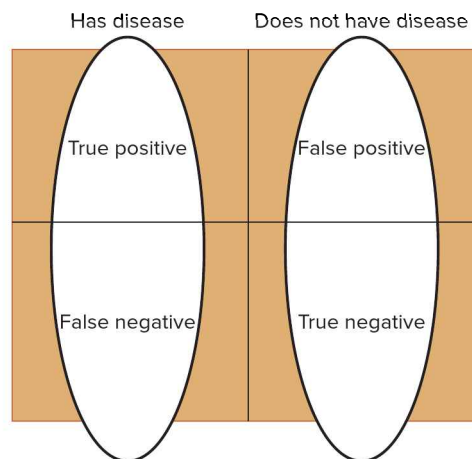


Figure 14.1 Sensitivity is measured among those individuals who have the disease. A great number of true positives and few false negatives lead to a high sensitivity (left ovoid). Specificity is measured among those who do not have the disease. In this case, a low number of false positives and a high number of true negatives generate a high specificity. Each of these numbers only represents half the diagnostic story.

is the probability that subjects with a positive screening test truly have the disease. Negative predictive value is the probability that subjects with a negative screening test truly don't have the disease.

Like sensitivity and specificity, positive and negative predictive values only capture a portion of the population (Fig. 14.2) and should not be used exclusively in clinical practice because they fail to tell the full story of the test's utility. Positive predictive values include only true positives and false positives. Negative predictive values include only false and true negatives.

PRACTICE PEARL

Very importantly, positive and negative predictive values are markedly influenced by the prevalence of disease in the population that is being tested. If we test in a high-prevalence setting, it is more likely that persons who test positive truly have disease than if the test is performed in a population with low prevalence, thus artificially supporting the diagnostic value of a positive finding on a test.

Likelihood ratio

A high positive likelihood ratio (LR+) influences post-test probability with a positive finding. A value of >1 rules in a diagnosis. A low negative likelihood ratio (LR-) influences post-test probability with a negative finding. A value closer to 0 is best and rules out the condition. Although others have suggested thresholds for independent decision making, likelihood ratios should be evaluated within the context of pre-test prevalence and in each unique clinical setting.

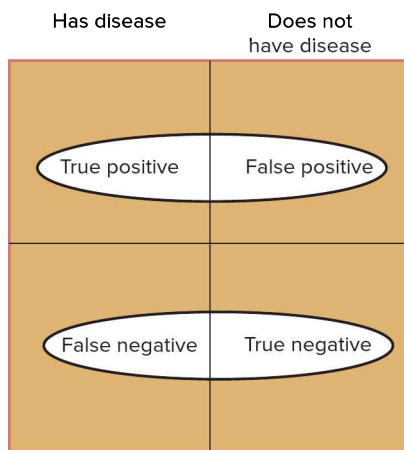


Figure 14.2 Positive and negative predictive values. These should be used when the clinician is trying to answer the question, 'The clinical screening test is positive. What is the likelihood that this patient truly has the condition?'

Both sensitivity and specificity are used to determine positive and negative likelihood ratios. LR+ is calculated using the formula $(\text{sensitivity})/(\text{100}-\text{specificity})$, whereas LR- is calculated using the formula $(\text{100}-\text{sensitivity})/(\text{specificity})$. In LR+ and LR- the full population of interest, including those with and without the disease of interest, is factored into the decision-making metrics.

This supports likelihood ratios as the most favourable metric for determining influence of post-test probability and for truly understanding the influence of a test on diagnosis accuracy. Likelihood ratios account for the prevalence, give further perspective, and allow for adjustments of test values in extreme cases of low or high prevalence. Some consistencies occur regardless of disease of interest. A higher pre-test probability will always improve the post-test probability. A lower pre-test probability will demand a strong LR+ (to rule in); conversely, a higher pre-test probability will require a lower LR- (to rule out) to notably alter the post-test probability.

Clinical utility

Clinical utility is a term used when evaluating the ability of the metrics to influence post-test probability (either in ruling out the condition or ruling it in) or just improving your likelihood of being correct. A nomogram is a routinely used mechanism to identify post-test probability. A nomogram has three scoring mechanisms, mounted vertically, left to right (Fig. 14.3).⁸

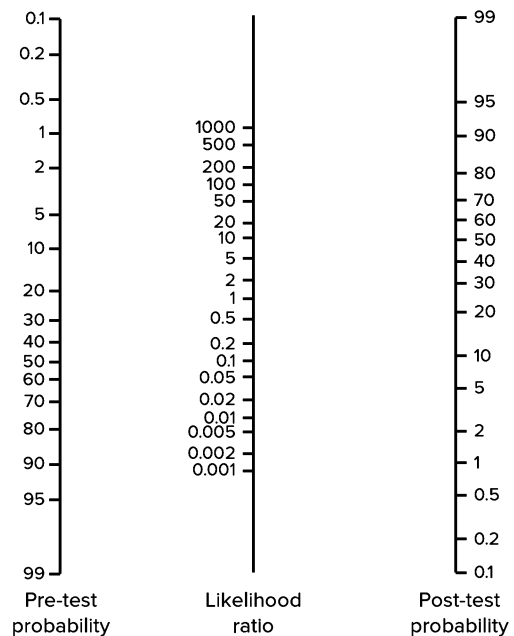


Figure 14.3 Fagan's nomogram⁸

REPRODUCED FROM FAGAN TJ. LETTER: NOMOGRAM FOR BAYES' THEOREM. N ENGL J MED. 1975;293(5):257

Applying Fagan's nomogram to the clinical setting: this tool is a key and underrated instrument for your career

Nomograms are an important learning mechanism to understand the interplay of pre-test probability, test metrics and post-test probability when differentially diagnosing. Consider the case of a patient with an anterior cruciate ligament (ACL) injury (Fig 14.4). Although ACL injuries are relatively uncommon per athlete exposure, the injuries make up a larger percentage of patients with knee problems in the sports medicine setting.⁹

If the percentage of individuals seen with an ACL injury in a busy sports medicine setting is 15%, one can assume that by chance, if the clinician 'guessed' ACL injury, he or she would be correct in 15% of cases. Since our goal is to improve post-test

probability, the clinician may use a Lachman's test to further evaluate the likelihood of the condition. In a meta-analysis, the Lachman's test had an LR+ of 4.5 and LR– of 0.2.¹⁰ Using the pre-test probability of 15% (based on numbers from the busy sports medicine setting) a positive Lachman's test would increase the post-test probability to 44% (green arrow, Fig. 14.4). In contrast, a negative finding on the Lachman's would result in a post-test probability of 4% (red arrow, Fig. 14.4).

Similar calculations can be performed for all conditions in all settings, with an understanding that pre-test prevalence will vary in different settings. Note that likelihood ratios are only as good as the influence they provide on post-test probability.

The left column represents the pre-test probability of the disease of interest with either no testing, based on the probability of occurrence in the practice environment of

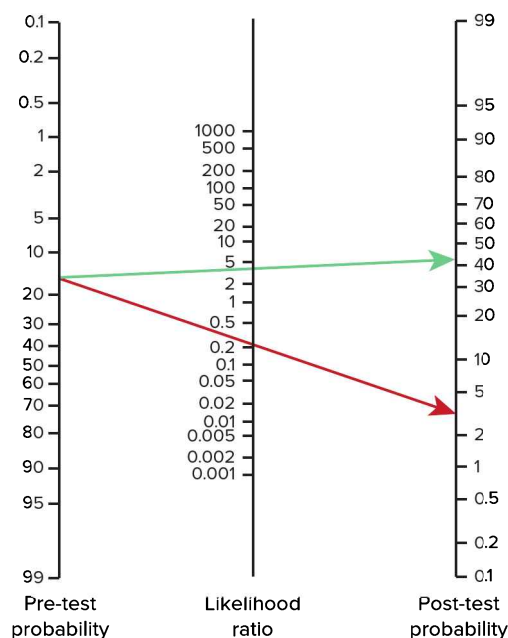


Figure 14.4 This illustrates the case where the pre-test probability (left) was 15%, LR+ (middle line) was 4.5 and the post-test probability (derived from those points if the test was positive, green arrow) was 44%. In practice, this means the clinician has increased his/her confidence about a diagnosis from 'relatively unlikely' to 'relatively likely'. This will affect the treatment plan and the conversation with the patient as to the treatment plan (see 'shared decision making' in Chapters 2 and 15.)

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the clinician (i.e. anterior cruciate ligament tears are more common in a sports medicine specialist's environment), or based on the data used to evaluate the sample. This value is scored 0% (low prevalence) to 99% (high prevalence). The middle column represents the inherent likelihood ratio (positive in the upper column, negative in the lower column) and is generally combined with the pre-test probability to determine post-test probability.

The final column (on the right of the nomogram) represents the post-test probability. This value is determined by running a line from the pre-test probability, through the middle column (test metric) to the post-test probability value (Fig. 14.4).

PRACTICE PEARL

Differential diagnosis is case dependent and several rules of thumb are useful.

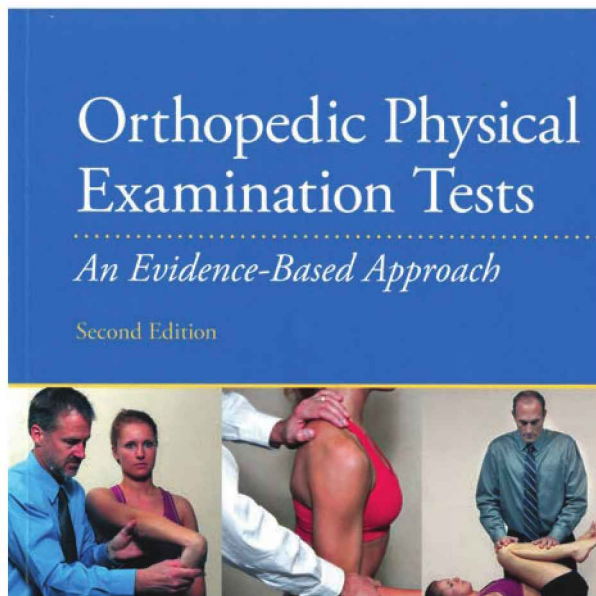
Tests will generally have high LR+ or a small LR– but rarely both. Consequently, the test metrics drive appropriate use: some should be used early to rule out a condition (those with small LR–) and some should be used later to confirm the presence of the condition (those with a large LR+). If the tests are used inappropriately, such as using a test with low LR– but also a low LR+, to rule in a condition, one is unlikely to accurately represent what the test is designed to do. Table 14.1 gives some examples of which tests are best for use early to 'rule out' a condition (small LR–) or late to 'rule in' a condition (large LR+).

For a comprehensive list of the best tests and measures, and each test's individual metrics, see the textbooks on physical examination by Professors Chad Cook and Eric Hegedus (Fig. 14.5)¹¹ and clinical assessment by Dr Michael Reiman (Fig. 14.5).¹² Both are essential resources.

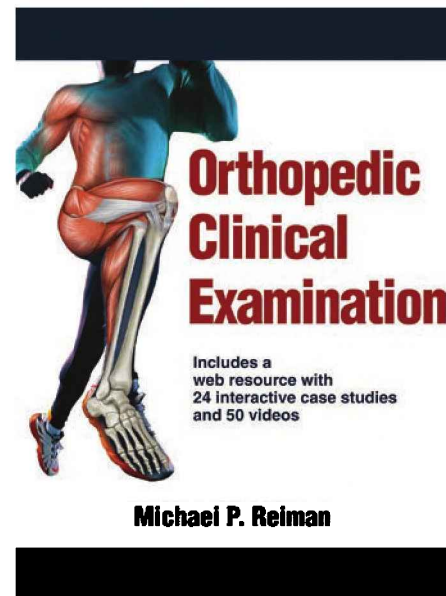
Table 14.1 Common tests and measures and the test's ability to 'rule in' or 'rule out' a condition¹¹

Test	Dominant test metric	Best use in clinical practice
Straight-leg raise for lumbar radiculopathy	A (negative) test has a small LR ⁻ and reduces to the post-test probability of lumbar radiculopathy	The test should be used early in the examination to rule out lumbar radiculopathy only
The External Rotation Lag sign for infraspinatus tears of the shoulder	A (positive) test has a high LR ⁺ and increases the post-test probability of an infraspinatus tear	The test should be used near the end of the examination to rule in an infraspinatus tear of the shoulder
The Hawkins–Kennedy test for impingement of the shoulder	A (negative) test has a low LR ⁻ and reduces the post-test probability of shoulder impingement	The test should be used early in the examination to rule out impingement of the shoulder
The patellar-pubic percussion test for the hip (fractured femur)	The test has both a low LR ⁻ and a high LR ⁺ and influences post-test probability with a negative and a positive	The test is a rare one that can rule in or rule out a hip fracture and can be used at all time points, respecting safety for the patient
The anterior drawer test for an ACL tear	A (positive) test has a high LR ⁺ and increases the post-test probability of an ACL tear	The test should be used near the end of the examination to rule in an ACL tear
The squeeze test for a syndesmosis injury at the ankle	The test does not exhibit a high LR ⁺ or a small LR ⁻ , thus it does not influence post-test probability	The test should not be used to rule in or rule out

With some adapted content from Cook C, Hegedus E. *Orthopedic Physical Examination Tests: An Evidence-Based Approach*. 2nd edn. Upper Saddle River NJ; Prentice Hall: 2013.



(a)



(b)

Figure 14.5 Two excellent resources to help you become a skilled clinician. (a) *Orthopedic Physical Examination Tests* by Professors Chad Cook and Eric Hegedus¹¹ (b) *Orthopedic Clinical Examination* by Dr Michael Reiman¹²

COOK, CHAD; HEGEDUS, ERIC, *ORTHOPEDIC PHYSICAL EXAMINATION TESTS: AN EVIDENCE-BASED APPROACH*, 2ND ED., ©2013. REPRINTED BY PERMISSION OF PEARSON EDUCATION, INC., NEW YORK, NEW YORK; REIMAN M.P., *ORTHOPEDIC CLINICAL EXAMINATION*, © 2016, REPRINTED BY PERMISSION OF HUMAN KINETICS, CHAMPAIGN, IL

THE FORMAL DIAGNOSTIC ASSESSMENT

Diagnosis relies on taking a careful history, performing a thorough physical examination and using appropriate investigations. There is a tendency for clinicians to rely too heavily on sophisticated investigations and to neglect their clinical skills.¹³ The goal of the clinical assessment when considering differential diagnosis is to improve post-test probability, thus providing the most appropriately aligned care to the athlete. During clinical assessment, all test findings have the capacity to influence post-test probability, including: 1. patient history and intake information; 2. observation; 3. the dedicated clinical (movement and performance) examination; and 4. special triage or confirmatory testing.

It is imperative to recognise that key features such as training history, nutrition, general health, work and leisure habits, past injury, equipment use, sports-specific demands, involvement in other sports and psychological features can markedly influence the diagnosis of a patient. These features paint a comprehensive picture of the state of the athlete's condition and the external factors that may drive future care needs.

It is well known that clinicians place too great a value on single testing mechanisms during differential diagnosis, and the tests do not have the capacity to discriminate in absence of these other important considerations. This includes ALL forms of tests, such as imaging, laboratory testing and so on. In essence, a good clinician evaluates all the information and places the test finding in context.

PRACTICE PEARL

Most clinicians overestimate the utility of special tests, assuming they provide more decision-making capacity than they do. Only use special tests in context to the rest of the clinical assessment.

The role of bias in influencing diagnostic metrics

Can you rely on published reports of test accuracy? Unfortunately not, because the design of a study can markedly influence its reported diagnostic accuracy. Recognising this, Whiting and colleagues created the Quality Assessment of Diagnostic Accuracy Studies tool (QUADAS) and subsequently QUADAS II.¹⁴ A number of researchers have recognised the influence of bias through use of QUADAS and its impact on elevating test metrics. Each study should be carefully analysed for risk of bias before considering the test's use in clinical practice. Some biases can elevate the value of accuracy metrics by four

PRACTICE PEARL

Diagnostic metrics from diagnostic accuracy studies that exhibit high risk of bias, regardless of where these studies were published, should not be incorporated into clinical practice. Bias markedly influences test metrics.

times the realistic amount, suggesting far greater utility of a test that may provide very little in clinical practice.

Challenges to making a diagnosis

Differential diagnosis is an imperfect science. For reasons beyond the limitations of testing, some diagnoses are extremely difficult to make. The following are examples of non-correctable challenges to diagnosis and deserve explanation.

- Verification bias occurs when the gold standard procedures (e.g. biopsy, surgery) are provided to only a select few (because of costs, effort and so on) and subsequently only reflect that select few. Unfortunately, the extra step needed to improve the accuracy limits the generalisability of the finding. Whether a more traditional population would yield the same result is unknown.¹⁵
- Syndromes refer only to the set of detectable (examination) characteristics (signs and symptoms) that are related to a defined diagnosis. Unlike diseases, which have hard imaging, clinical or laboratory findings, syndromes are purely based on numerous intangible features (e.g. thoracic outlet syndrome), increasing the likelihood that two individuals can be diagnosed with the same problem, despite exhibiting notably different clinical findings.
- Condition severity is a known mediator to diagnostic accuracy. For example, a superior labral anterior posterior (SLAP) lesion can be categorised into four distinct groups using Snyder's classification.¹⁶ Type I and type II classifications exhibit clinical features that are strikingly similar to a rotator cuff tear, impingement, bursitis and so on, and these classifications are notably difficult to differentiate in clinical practice in comparison to the more severe classifications of type III and IV.
- Incorporation bias occurs when the test is part of the diagnosis. An example is patellofemoral pain syndrome which is a diagnosis based on anterior knee pain and challenges with functional activities. Often, the functional activities are testing for their predictability; despite that, these are part of the diagnosis itself. The main problem with incorporation bias is its overestimating diagnostic accuracy.

FINAL THOUGHTS AND GUIDANCE

There are a number of considerations one must make that go beyond diagnostic metrics and clinical tests and measures. First, as David Matcher implies, few tests markedly change clinical practice for the good of society.¹⁷ One can argue that a minority of our clinical tests and measures are superfluous and do little in assisting in decision making for the long-term good of our athlete. This is likely why we have a proliferation of newly created clinical tests for diagnosis of the shoulder, the sacroiliac joint and other problematic regions.

It is imperative that differential diagnosis include a careful discussion with the athlete and the implications of the finding. Shared decision making is necessary for both parties to consider which treatment is appropriate for the condition. By being aware of the limitations highlighted in this chapter, clinical assessment and judicious use of imaging provides a great foundation for the treatment plan. In Chapter 15, we share some practical clinical approaches you can use to maximise your chances of arriving at the most likely diagnosis! The great clinician combines science and art.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

